

Report R04 — Glioma Molecular Epidemiology and Prognosis

Two cohorts, ~2,000 patients, no imaging. Includes a correction to the comparison with R02.

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Summary

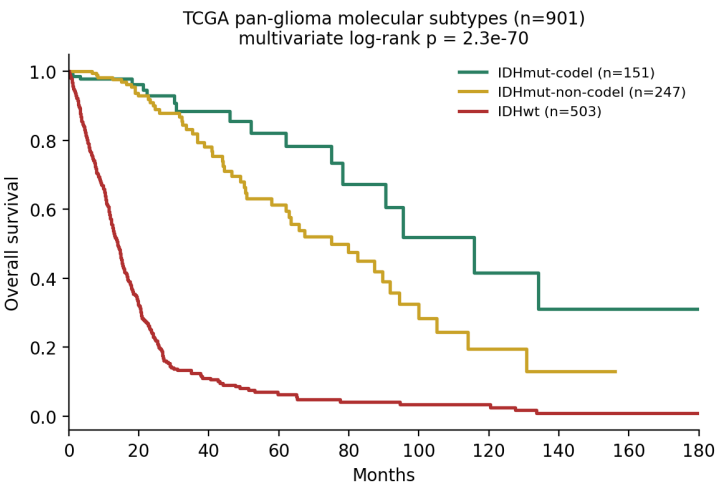
Across TCGA pan-glioma ($n = 1,040$ with survival) and the MSK panel cohort ($n = 923$), the molecular markers behave exactly as established biology predicts, and the pipeline recovers them at overwhelming significance. IDH mutation carries a median survival of 89.7 months versus 14.0 for wild-type ($p = 5.7 \times 10^{-71}$). The three-tier CNS5 subtype stratification separates cleanly: codeleted 115.8 months, non-codeleted 75.1, IDH-wildtype 14.0 ($p = 2.3 \times 10^{-70}$).

The headline finding is a correction, not a claim. A naive reading of these data says the molecular panel (C-index 0.797) far outperforms imaging (0.602 in R02). *That comparison is invalid*, and the audit caught it before this report was written. TCGA spans grades 2–4 while UPENN-GBM is glioblastoma only, and IDH status largely *is* the grade distinction. Restricting TCGA to glioblastoma — the like-for-like comparison — the molecular panel collapses to **0.541–0.561**, below the **0.602** that imaging achieved.

1 Results

Positive control and stratification

Group	<i>n</i>	Median OS (months)	Test
IDH mutant	919	89.7	$p = 5.7 \times 10^{-71}$
IDH wild-type		14.0	
IDHmut-codeleted	151	115.8	$p = 2.3 \times 10^{-70}$
IDHmut-non-codeleted	247	75.1	
IDH wild-type	503	14.0	



Marker co-occurrence — the CNS5 variables are not independent

Pair	Odds ratio	p	q (BH)	Reading
IDH $\times$ TERT promoter	0.186	7.6×10^{-10}	1.1×10^{-9}	mutually exclusive
IDH $\times$ MGMT	17.11	3.5×10^{-58}	1.1×10^{-57}	strongly co-occurring
TERT $\times$ MGMT	1.05	0.89	0.89	independent

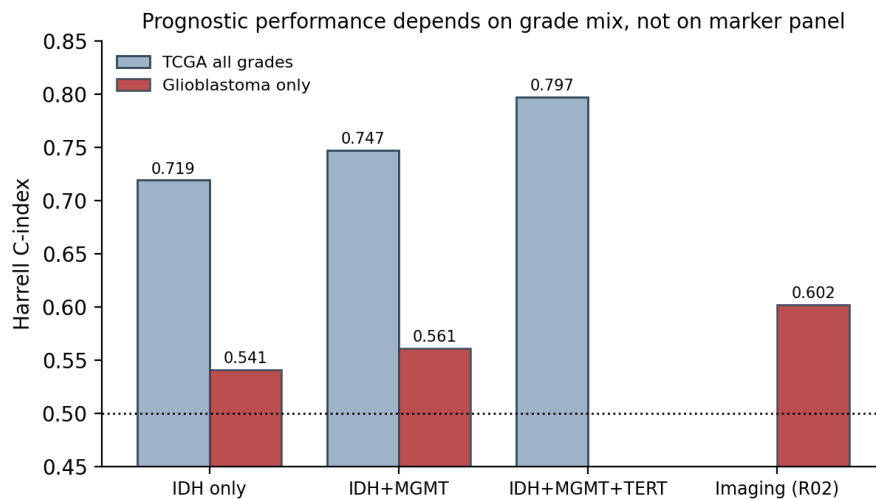
Both non-null results match known biology: IDH-mutant and TERT-promoter-mutant tumours use largely alternative routes, and IDH-mutant gliomas are overwhelmingly MGMT-methylated (the G-CIMP phenotype). This is a second, independent confirmation that the data and joins are sound.

Independent replication in the MSK cohort ($n = 923$)

Marker	n positive	HR	p	C-index
TERT promoter mutation	548	1.69	2.8×10^{-6}	0.591
CDKN2A homozygous deletion	283	2.37	4.3×10^{-14}	0.588
Both	—	—	—	0.627

2 The correction: prognostic performance tracks grade mix, not marker panel

Predictor	TCGA all grades	Glioblastoma only	n (GBM)
IDH only	0.719	0.541	470
IDH + MGMT	0.747	0.561	356
IDH + MGMT + TERT	0.797	not estimable	26
Imaging alone (R02, UPENN-GBM)	—	0.602	574



Within glioblastoma, the standard molecular panel carries little prognostic information (0.541–0.561) while imaging carries more (0.602). The panel’s apparent strength across all grades is real but largely tautological: IDH status is close to a proxy for whether the tumour is a low-grade glioma or a glioblastoma, and that distinction dominates survival.

Cross-validation optimism was also checked and is modest — 0.719 falls to 0.707 and 0.797 to 0.759 under 5-fold CV — so it is the cohort composition, not overfitting, that drives the

difference.

3 What R02 and R04 say together

1. Molecular markers are outstanding at separating glioma *types* and therefore survival across grades. Nothing here challenges that.
2. Within glioblastoma specifically, they add little prognostically — and imaging adds more.
3. R02 showed the imaging signal is independent of age ($p = 1.6 \times 10^{-9}$). R04 now shows it is also not simply recapitulating the molecular panel, because that panel performs worse in the same disease setting.
4. R01 remains consistent: imaging does not encode MGMT status, and MGMT itself adds essentially nothing within glioblastoma ($0.719 \rightarrow 0.747$ across grades, $0.541 \rightarrow 0.561$ within GBM).

That is a coherent, defensible position for the glioma arm: **imaging is worth studying prognostically in glioblastoma precisely because the molecular panel is weak there.**

4 Limitations

- The imaging and molecular results come from *different cohorts* (UPENN-GBM vs TCGA). They are matched on disease but not on population, treatment era or follow-up. A within-cohort head-to-head would need molecular markers on the UPENN patients — IDH1 and MGMT are available there and this is the obvious next analysis.
- TERT is available on only 311 TCGA patients and just 26 glioblastomas, so the three-marker panel cannot be evaluated within GBM at all.
- C-indices for the panels are from Cox models on binary markers, not a tuned learner.
- TCGA treatment is heterogeneous and pre-dates current standards.

5 Audit

Check	Verdict	Finding
Positive control	Recovered	IDH $p = 5.7 \times 10^{-71}$, correct direction
Cohort comparability	FAILED as first framed	TCGA spans grades 2–4; UPENN is GBM only. Comparison corrected in §3
CV optimism	Modest	$0.719 \rightarrow 0.707$; $0.797 \rightarrow 0.759$
Co-occurrence sanity	Matches biology	IDH/TERT exclusive, IDH/MGMT co-occurring
Independent replication	Yes	TERT and CDKN2A both prognostic in MSK, $n = 923$
Gates	4 / 4 PASS	After one documented failure, see below

A gate failed on the first attempt: expected 1,046 survival records, observed 1,040. Six patients have `OS_MONTHS = 0`, which a Cox model cannot use. The expectation was too tight; it was corrected to 1,040 with the reason recorded in source, and the failed run is retained as `gates_FAILED_first_attempt.json`.

6 How to verify

`evidence/` contains `config.yaml` (hash `18ea8aa0...`, written before any outcome was read), `gates.json`, the failed run's gates, `results.json` with every statistic, `tcga_survival.csv`, `inputs.json` (file hashes), `MANIFEST.sha256`, `env.txt`, and both scripts.

Run `runs/20260806T235340Z-R04glioma-3f8bece`. Data: cBioPortal `lgggbm_tcga_pub` (Broad GDAC/NIH terms) and `glioma_mskcc_2019` (ODbL). No patient-level identifiers reproduced.